

CASE STUDY

10 Marks

April/May 2025

- Q Solve the following case study with appropriate explanation
A healthcare organization lost patient data due to an accidental deletion of their AWS hosted database.
How can disaster recovery planning help avoid such losses? what AWS tools and strategies can ensure quick data recovery after accidental deletions?

⇒ Introduction

A healthcare organization lost important patient data because their AWS hosted database was accidentally deleted. Patient records contain medical history, reports, prescriptions, & treatment details.

Loss of this data creates serious problem because hospital work may stop, patient treatment may be delayed & trust of patient may be affected.

Since healthcare data is sensitive, quick recovery is important.

cause of Problem

The main reason for this issue was accidental deletion of the database and the absence of proper disaster recovery plan. The organization may not have enabled regular backups, automated snapshots, or recovery option like Point-in-Time Recovery.

Another reason could be poor IAM configuration. If many users have permission to delete or modify the database, the chance of accidental deletion becomes higher. Lack of Multi-Factor Authentication (MFA) and weak monitoring system also increase the risk.

Without proper logging & alerts, the organization

may not know immediately when deletion happens, which delays recovery & increases data loss.

Solution / Preventive Measures

1. Enable Automated Backups.

The organization should enable AWS RDS automated backup so that database backup are created regularly without manual work. This helps in restoring lost data easily. It also reduces the risk of permanent data loss during accidental deletion or system failure.

2. Use Manual Snapshots

Manual snapshots should be taken before major updates or changes in the database. These snapshots help restore the database if accidental deletion happens. They provide an extra safety layer before performing important operations.

3. Enable Point-in-Time Recovery (PITR)

Point-in-Time Recovery allows restoring the database to a specific time before the deletion happened. This reduces data loss & helps quick recovery. It is especially useful when recent transactions need to be recovered accurately.

4. Use AWS Backup

AWS Backup Service helps to manage all backups from one place. It can schedule backups automatically & monitor backup status easily.

This improves backup management & ensures regular data protection without manual efforts.

5. Apply IAM Security

IAM should be used to control who can delete or modify the database. Only authorized users should get access by following the principle of least privilege. This reduces the chances of accidental deletion by unauthorized users.

6. Enable MFA

Multi-Factor Authentication (MFA) should be enabled for admin users to provide extra security & prevent unauthorized actions. Even if a password is stolen, MFA helps to stop unauthorized access.

7. Use Multi-AZ Deployment

Multi-AZ deployment keeps a copy of data in another availability zone. This improves availability & helps faster recovery during failure. It also reduces downtime and keeps hospital services running smoothly.

8. Monitoring with CloudTrail & CloudWatch

AWS CloudTrail records all activities like deletion events, & CloudWatch sends alerts for suspicious action. This helps in quick response & recovery. It also helps administrators identify the cause of problem & take preventive action quickly.

Conclusion

By using Proper disaster recovery planning, regular automated backups, manual snapshots, point-in-Time Recovery, IAM security, MFA, Multi-AZ deployment & monitoring tools like CloudTrail & CloudWatch, the healthcare organization can protect important patient data effectively.

These AWS Tools helps to reduce the risk of accidental deletion, improve data security & ensure quick recovery during emergencies. This increases system reliability, patient trust & smooth hospital operations.

Nov / Dec 2025

10 Marks

- Q Solve the following case study with appropriate explanation
A financial services company faced a data breach when their AWS RDS backups were publicly exposed due to misconfiguration of S3 Bucket Policies. How can IAM policies and Encryption improve backup security?

⇒ Introduction

A financial services company faced a data breach because their AWS RDS backups were publicly exposed due to misconfiguration of S3 bucket policies. These backups contained sensitive customer information such as account details, transaction records & financial documents.

If such confidential data become accessible to unauthorized users, it can lead to financial fraud, data theft, legal issues, & loss of customer trust. Since financial data is highly sensitive, strong backup security is very important.

This type of data breach can also affect the company's reputation in the market & may result in heavy financial penalties due to non-compliance with security regulations. Quick action is necessary to prevent further misuse of customer information.

cause of Problem

A) Misconfigured S3 Bucket Policy

The main reason for the problem was incorrect S3 bucket policy setting, which allowed public access to the backup files. This made sensitive RDS Backup data accessible to unauthorized users.

B) Weak IAM Policies

Proper IAM (Identity & Access Management) policies were not applied. Too many users may have had unnecessary permissions to access, modify or delete backup files, which increased security risk.

C) Lack of Encryption

The backup files were not properly encrypted using AWS KMS or S3 Server Side encryption. Without encryption, sensitive financial data can be easily read if an unauthorized access happens.

D) No Monitoring & Regular Security checks

AWS CloudTrail, S3 access logs, & regular policies. Policy reviews were not properly maintained. Because of this suspicious activities & configuration mistakes remained unnoticed for long time.

Solution / Preventive Measures

1. Apply IAM Least Privilege Policy

The company should use IAM policies based on the principle of least privileges. This means users should get only the minimum permissions required for their job. This reduces the chance of unauthorized access to backup files.

2. Block Public Access for S3 Buckets

The "Block Public Access" feature should be enabled for all S3 buckets storing RDS backups. This preventive accidental public exposure of sensitive backup data. It adds an important security layer.

for cloud storage.

3. Use IAM Roles Instead of Access Key

IAM Roles should be used instead of sharing access key or passwords. Roles provide temporary & secure access to AWS resources. This improves security & reduces the risk of credential misuse.

4. Enable Multi-Factor Authentication (MFA)

MFA should be enabled for admin users & important accounts. It adds an extra verification step during login. Even if a password is stolen, unauthorized users cannot easily access the system.

5. Encrypt RDS Backups Using AWS KMS

AWS RDS backups & snapshots should be encrypted using AWS KMS (Key Management Services). Encryption converts data into unreadable form for unauthorized users. This protects sensitive financial data even if backup files are exposed.

6. Enable Server-Side Encryption in S3

The S3 bucket storing backups should use server-side encryption like SSE-KMS or SSE-S3.

This ensures that all backup files stored in S3 remain protected automatically. It improves overall backup security.

7. Restrict Access to Encryption Key

Access to encryption key should also be controlled using

IAM and kms policies. only trusted user & applications should be allowed to use decryption keys. This prevent misuse of sensitive data.

8. Monitoring with CloudTrail & S3 Access Logs

AWS CloudTrail and S3 access logs should be enabled to monitor all access activities. These tools help data to detect suspicious access attempts & policy changes. Quick alerts help administrator respond faster and prevent major security incidents. Regular monitoring also helps in auditing & maintaining security standards.

Conclusion

By using strong IAM policies, blocking public access, enabling MFA, applying encryption through AWS KMS & monitoring with CloudTrail, the company can protect AWS RDS backups from unauthorized access.

These security measures reduce the risk of data breach, protect customer financial information & improve trust & compliance in the organization.